

LAB: Digital Image Processing

Lab on Morphological Image processing

Objective:

- I. To understand and apply the basic morphological image processing operations such as erosion/dilation using OpenCV in python for shape-based image analysis.

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Introduction to Morphological IP?

- Morphological image processing is a **set of operations** that process images based on shapes.
- These operations are most commonly applied on binary images (black and white), but can also be extended to grayscale images.
- Morphological operations use a small matrix called a structuring element (kernel), which slides over the image and modifies pixels based on specific rules.

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Structuring Element (Kernel)?

- A structuring element is a small shape such as:
- 3×3 square
- Cross
- Circle (approx)
- Custom shapes
-

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Basic Morphological Operations

- *Erosion*
- Removes pixels on object boundaries
- Makes objects **thinner**
- Removes small noise
- Useful for separating connected objects
- Formula:

Output pixel = **minimum** value under the kernel

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Basic Morphological Operations

- **Dilation**
- Adds pixels to object boundaries
- Makes objects **thicker**
- Fills small holes
- Useful for connecting broken parts
- Formula:
Output pixel = **maximum** value under the kernel

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Basic Morphological Operations

- **Opening = Erosion $\rightarrow$ Dilation**
- Removes small white noise
- Keeps larger shapes safe
- **Closing = Dilation $\rightarrow$ Erosion**
- Fills small black holes
- Smooths object boundaries

Required Libraries

```
import cv2  
import numpy as np  
import matplotlib.pyplot as plt
```

```
[1]: import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
[2]: img = cv2.imread('img2.jpg', 0)    # Load grayscale image
plt.imshow(img, cmap='gray')
plt.title("Original Image")
plt.axis('off')
plt.show()
```

Original Image




```
[3]: kernel = np.ones((5, 5), np.uint8)   # 5x5 square kernel
     kernel
```

```
[3]: array([[1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1],
           [1, 1, 1, 1, 1]], dtype=uint8)
```

```
[4]: erosion = cv2.erode(img, kernel, iterations=1)

plt.figure(figsize=(8,4))
plt.subplot(1,2,1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1,2,2); plt.imshow(erosion, cmap='gray'); plt.title('Erosion'); plt.axis('off')
plt.show()
```

Original



Erosion



```
[5]: dilation = cv2.dilate(img, kernel, iterations=1)

plt.figure(figsize=(8,4))
plt.subplot(1,2,1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1,2,2); plt.imshow(dilation, cmap='gray'); plt.title('Dilation'); plt.axis('off')
plt.show()
```

Original



Dilation



```
[6]: opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)

plt.imshow(opening, cmap='gray')
plt.title("Opening")
plt.axis('off')
plt.show()
```

Opening



```
[7]: closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)

plt.imshow(closing, cmap='gray')
plt.title("Closing")
plt.axis('off')
plt.show()
```

Closing



```
[8]: gradient = cv2.morphologyEx(img, cv2.MORPH_GRADIENT, kernel)

plt.imshow(grad, cmap='gray')
plt.title("Morphological Gradient")
plt.axis('off')
plt.show()
```

Morphological Gradient



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Student Exercises

1. Perform dilation with kernel sizes (3×3) , (7×7) , (11×11)
2. Perform erosion 2 and 3 iterations
3. Apply Opening on a noisy image
4. Apply Closing on an image with holes
5. Extract edges using morphological gradient